10.8 Three Stars in a Triangle

WORKSHEET

The question

STEP I 1999, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q11

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself For a star to move in a circle of radius R with angular velocity ω, what must the total force on it be?

STAGE 1 The geometry

Ask yourself How far is each corner of an equilateral triangle of side a from its centre?

STAGE 2 The force on one star

Ask yourself Each of the other two stars pulls with force $\frac{\gamma m^2}{a^2}$. What is the angle between the two pulls? What is their resultant, and in which direction does it point?

STAGE 4 A fourth star at the centre

Ask yourself A star of mass λm at the centre is at distance $\frac{a}{\sqrt{3}}$ from each corner. What extra force does it exert on each corner star? Does the centre of mass move? Does the central star feel any net force?

What's really going on

Kepler's law in disguise

Write the answer as $\omega^2 a^3 = \gamma(3m)$. For two bodies in circular orbit a distance a apart, Kepler's third law (an exercise in Section 10.6) says $\omega^2 a^3 = \gamma(m_1 + m_2)$, the total mass. The three stars obey the same law, with the total mass $3m$. In fact Lagrange showed that the equilateral solution exists for *any* three masses, not just equal ones: the triangle rotates about the centre of mass with $\omega^2 a^3 = \gamma(m_1 + m_2 + m_3)$. Euler had found a second family a few years earlier, with the three bodies on a rotating straight line. These five configurations (three collinear, two triangular, up to labelling) are the only ones in which three bodies keep the same shape.

Lagrange points

Take m_1 to be the Sun, m_2 a planet, and m_3 a tiny asteroid. The equilateral solution says the asteroid can sit at the third corner of an equilateral triangle with the Sun and the planet, ahead of or behind the planet in its orbit. These are the *Lagrange points* L_4 and L_5; Euler's collinear solutions give L_1, L_2 and L_3. Jupiter has over ten thousand known "Trojan" asteroids clustered around its L_4 and L_5 points. The James Webb Space Telescope orbits near the Sun–Earth L_2 point, 1.5 million km beyond the Earth, where it can keep the Sun, Earth and Moon all behind its sunshield.

Stability

Would the Starmakers' triangle last? For equal masses, no: the slightest disturbance grows, and the stars fly apart or collide. Edward Routh showed in 1875 that Lagrange's triangle is stable only if one mass dominates, precisely when

$$27(m_1 m_2 + m_2 m_3 + m_3 m_1) < (m_1 + m_2 + m_3)^2.$$

For the Sun and Jupiter this holds comfortably, which is why the Trojans survive. For three equal masses, $27 \times 3m^2 > 9m^2$, so the configuration is unstable.

Poincaré's mistake and the figure eight

In 1887 King Oscar II of Sweden offered a prize for a solution of the many-body problem. Henri Poincaré won it in 1889 with a long memoir on the three-body problem, but while the memoir was being printed an editor found a gap. In correcting it, Poincaré discovered that some solutions behave with "a complexity so astonishing that I cannot even attempt to draw it": the first glimpse of chaos. The printed copies were recalled and destroyed, at Poincaré's expense. New special solutions are still being found: in 1993 Cristopher Moore discovered numerically, and in 2000 Alain Chenciner and Richard Montgomery proved, that three equal masses can chase each other round a figure-eight curve. Thousands more periodic orbits have since been found by computer.

Try this next

1. **Four stars in a square.** Four stars of mass m are placed at the corners of a square of side a. Show that they can rotate rigidly about the centre with $\omega^2 = \frac{(4+\sqrt{2})\gamma m}{2a^3}$.

2. **Two equal stars and a planet.** Two stars of mass M orbit their centre of mass, a distance a apart. A planet of negligible mass sits at the centre. Is it in equilibrium? Is it stable to small displacements along the line joining the stars? Perpendicular to it?

3. **Unequal masses.** Three stars of masses m_1, m_2, m_3 are at the corners of an equilateral triangle of side a, rotating about their centre of mass G. Show that the gravitational force on m_1 is $\frac{\gamma m_1(m_1+m_2+m_3)}{a^3}\overrightarrow{P_1G}$, where P_1 is its position. (*Hint:* write the forces as vectors and use the definition of the centre of mass.) Deduce Lagrange's result.

4. n **stars.** n equal stars sit at the corners of a regular n-gon inscribed in a circle of radius R. Find ω^2 for rigid rotation as a sum over k of $\frac{1}{\sin(k\pi/n)}$.

Further reading

- **June Barrow-Green, *Poincaré and the Three Body Problem*** (American Mathematical Society). The story of the prize and Poincaré's discovery of chaos.
- **Richard Montgomery, "A new solution to the three-body problem"**, *Notices of the American Mathematical Society* 48 (2001). The figure eight, explained by one of its discoverers; free online.
- **Florin Diacu & Philip Holmes, *Celestial Encounters*** (Princeton University Press). A popular history of the three-body problem and chaos.

The full guided solution is in Section 10.8:
https://beyondstep.keithmcnulty.org/ch10/three-stars.html